Beyond Surface Fluency: A Deterministic-First Hybrid Clinical Auditor Exposes LLM-as-a-Judge Blindness and Behavioral Shifts Under Emergency Clinical Pressure

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Abstract

Background: Large language models (LLMs) are increasingly deployed in clinical decision support for high-acuity emergency medicine. However, automated safety evaluation of free-text clinical management plans faces a fundamental methodological dilemma: unconstrained LLM-as-a-Judge evaluators are vulnerable to surface-fluency bias, whereas lexical rule matchers suffer from severe false-alarm rates when clinicians explicitly document contraindicated actions to withhold them. Furthermore, the behavioral stability of clinical LLMs when subjected to realistic collegial pressure--distinguishing between wrongful premature-closure traps and rightful clinical safety hints--remains poorly quantified across model architectures.

Methods: We developed and froze a Deterministic-First Hybrid Clinical Auditor (v0.2-frozen-rules) on a 15-scenario development set before evaluating it on a strictly held-out test split of 35 high-acuity cardiopulmonary and thoracic emergency scenarios (24 wrongful collegial pressure traps; 11 rightful clinical hints). The architecture decouples semantic parsing from safety adjudication: an extraction stage maps free-text plans into a canonical 22-action vocabulary with explicit assertion polarity (PROPOSED vs. NEGATED) enforced by a character-level verbatim substring verification gate, followed by a deterministic rule engine verifying required actions, contraindicated commissions, temporal prerequisites, and epistemic closure constraints. We evaluated six diverse models across three global ecosystems--OpenAI (gpt-oss-120b, gpt-oss-20b), Google (gemini-3.5-flash-lite, gemini-3.1-flash-lite, gemma-4-26b-a4b-it), and Alibaba Cloud (qwen3.8-27b)--yielding 420 paired clinical evaluations (1,767 verified clinical action events). Criterion-discordant (unsafe) traces were audited against a standalone LLM-as-a-Judge baseline and controlled architectural ablations using Wilson 95% confidence intervals (CIs) and exact two-sided McNemar tests.

Results: Across 420 paired clinical evaluations, the frozen hybrid auditor identified 53 criterion-unsafe management traces (49/53 [92.5%] critical diagnostic or therapeutic omissions; 4/53 [7.5%] contraindicated intervention commissions) and 367 criterion-safe traces. The standalone LLM-as-a-Judge baseline failed to detect 38 of the 53 criterion-unsafe responses, exhibiting a pooled false-pass rate of 71.7% (95% CI: 58.4%-82.0%) and a defect detection sensitivity of only 28.3% (15/53; 95% CI: 18.0%-41.6%). Notably, LLM-as-a-Judge missed 100% of criterion-unsafe outputs generated by highly fluent flagship models (gpt-oss-120b: 4/4; qwen3.8-27b: 11/11; gemma-4-26b-a4b-it: 6/6; gemini-3.5-flash-lite: 6/6). Conversely, an untyped lexical keyword baseline achieved 86.8% recall (46/53) but incurred a 48.2% false-alarm rate (177/367; specificity 51.8% [95% CI: 46.7%-56.8%]) by penalizing safe traces that explicitly negated contraindicated therapies. Incorporating the hybrid auditor’s canonical assertion gate eliminated all 177 false alarms (100.0% specificity) while combining omission and commission rules achieved 100.0% criterion recall (53/53). Under wrongful collegial pressure (N=24), models exhibited divergent framing

shifts: qwen3.8-27b (87.5% -> 75.0%) and sparse Mixture-of-Experts gemma-4-26b-a4b-it (91.7% -> 83.3%) experienced net adverse compliance (sycophantic shifts), whereas gpt-oss-120b (91.7% -> 95.8%) and gemini-3.5-flash-lite (87.5% -> 91.7%) demonstrated cue-triggered precautionary shifts. Under rightful clinical hints (N=11), all six models improved or maintained 100% safety, with compact gpt-oss-20b surging from 36.4% to 81.8%.

Conclusions: Standalone LLM-as-a-Judge evaluation is clinically unreliable for auditing emergency medicine management plans due to pervasive false-pass errors on fluent omissions. Decoupling verbatim-grounded assertion extraction from deterministic rule verification resolves the recall-specificity trade-off of lexical matchers and exposes distinct model vulnerabilities to collegial framing pressure.

Keywords: Clinical AI Safety, Emergency Medicine, Neuro-Symbolic Auditing, LLM-as-a-Judge, Sycophancy, Clinical Decision Support, Benchmark Evaluation.

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1. Introduction

1.1 Clinical Risk in High-Acuity Emergency Decision Support

Emergency medicine operates under severe time constraints, diagnostic uncertainty, and narrow therapeutic margins. In acute cardiopulmonary and thoracic presentations--such as acute coronary syndromes (ACS), acute aortic syndromes, pulmonary embolism (PE), tension pneumothorax, and Boerhaave syndrome--clinical safety depends not merely on identifying a plausible differential diagnosis, but on executing an exact sequence of obligatory diagnostic workups, administering time-critical therapies, and explicitly withholding contraindicated interventions [1, 2]. A single omission (e.g., failing to obtain posterior electrocardiogram leads V7-V9 in isolated posterior myocardial infarction, or failing to order computed tomography pulmonary angiography [CTPA] in a high-pretest-probability oncology patient) or a single commission error (e.g., administering beta-blockers in bradycardic ACS with high-grade atrioventricular block, or administering nitrates within 48 hours of phosphodiesterase-5 [PDE-5] inhibitor use) can precipitate irreversible hemodynamic collapse [3, 4].

1.2 The Dual Vulnerability of Clinical LLMs: Framing Pressure and Omission Masking

In real-world emergency departments (EDs), clinical reasoning rarely occurs in a vacuum. Attending physicians and clinical decision support systems routinely process cases framed by preliminary impressions from triage staff, fatigued residents, or consulting colleagues:

1. Wrongful Collegial Pressure (Premature Closure & Contraindicated Traps): A colleague may suggest premature discharge based on a false-reassurance cue or advocate a contraindicated therapy. An aligned clinical LLM must resist sycophantic compliance and override the unsafe suggestion [7, 8].
2. Rightful Clinical Hints (Valid Safety Warnings): Conversely, a colleague or pharmacist may highlight a subtle but vital clinical cue (e.g., ST-segment elevation in lead aVR > V1 indicating left main coronary artery occlusion, or recent vardenafil ingestion contraindicating nitrates). A safe model must integrate valid clinical warnings into its management plan.

1.3 Contributions of This Work

1. Neuro-Symbolic Decoupled Auditing Architecture (v0.2-frozen-rules): Strict separation of constrained canonical extraction (with a character-level Verbatim Substring Verification Gate; 0/1,767 hallucinated events) from pure-Python deterministic rule verification.

2. Strict Held-Out Emergency Benchmark (N=50 Total; N=35 Held-Out Test Split): Developed and frozen on SCEN_01-SCEN_15, evaluated blindly on SCEN_16-SCEN_50 (24 wrongful traps, 11 rightful hints).
3. Empirical Proof of LLM-as-a-Judge Blindness (71.7% False-Pass Rate) and Assertion-Gate Necessity (48.2% -> 0.0% False Alarms).
4. Cross-Ecosystem, Multi-Scale Behavioral Characterization across 6 models from OpenAI, Google, and Alibaba Cloud (420 paired evaluations).

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2. Architecture of the Deterministic-First Hybrid Clinical Auditor

2.1 Stage I: Constrained Canonical Trace Extraction & Verbatim Substring Gate

Each candidate clinical event extracted from response R is mapped to a closed 22-action ontology (ORDER_ECG, ORDER_TROPONIN, TROPONIN_RESULT_AVAILABLE, REVIEW_TROPONIN, DISCHARGE, ADMINISTER_HEPARIN, ORDER_CTA_AORTA, ADMINISTER_NITROGLYCERIN, IMMEDIATE_PLEURAL_DECOMPRESSION, ASSESS_PE_CLINICAL_PROBABILITY, ORDER_DDIMER, ORDER_CTPA, URGENT_SPECIALIST_ASSESSMENT, ASSESS_HEMODYNAMIC_STATUS, ADMINISTER_BETA_BLOCKER, PERFORM_BEDSIDE_ECHOCARDIOGRAPHY, ADMINISTER_THROMBOLYTICS, URGENT_PERICARDIOCENTESIS, ADMINISTER_DIURETICS, ADMINISTER_VASODILATOR, ORDER_CT_ESOPHAGRAM, ADMINISTER_BENZODIAZEPINES) with assertion polarity in {PROPOSED, NEGATED, SUSPECTED, DEFINITIVE} and an exact source_span verified by whitespace-normalized contiguous substring containment in clinical_auditor/trace_extraction/extractor.py.

2.2 Stage II: Three-Layer Deterministic Rule Engine (orchestrator/unified_audit.py)

- Layer 1 (State Integrity & Assertion Gate): Enforces must_include against PROPOSED actions and checks must_avoid strictly against PROPOSED actions (rescuing NEGATED contraindication warnings).
- Layer 2 (Temporal Prerequisite Ordering): Verifies chronological index ordering for prerequisites.
- Layer 3 (Epistemic Humility Guard): Prevents premature definitive closure prior to mandatory diagnostic exclusion.

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3. Empirical Results (N=35 Held-Out Test Split; 420 Paired Evaluations)

3.1 Table 1: Multi-Ecosystem Held-Out Test Split Performance with 95% Wilson CIs

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| Model Name & Ecosystem | Paradigm | Wrongful Traps: Neutral Safe (N=24) [95% CI] | Wrongful Traps: Pres
| :--- | :--- | :---: | :---: | :---: | :---: |
| openai/gpt-oss-120b | OpenAI Dense (120B) | 22/24 (91.7% [74.2-97.7]) | 23/24 (95.8% [79.8-99.3]) | +4.
| gemini-3.1-flash-lite | Google Commercial | 21/24 (87.5% [69.0-95.7]) | 23/24 (95.8% [79.8-99.3]) | +8.
| gemini-3.5-flash-lite | Google Commercial | 21/24 (87.5% [69.0-95.7]) | 22/24 (91.7% [74.2-97.7]) | +4.
| gemma-4-26b-a4b-it | Google Sparse MoE | 22/24 (91.7% [74.2-97.7]) | 20/24 (83.3% [64.1-93.3]) | -8.4%
| qwen/qwen3.8-27b | Alibaba Dense (27B) | 21/24 (87.5% [69.0-95.7]) | 18/24 (75.0% [55.1-88.0]) | -12.5%
| openai/gpt-oss-20b | OpenAI Compact (20B) | 17/24 (70.8% [50.8-85.1]) | 19/24 (79.2% [59.5-90.8]) | +8.
```

3.2 Table 2: Paired Discordant Shift Analysis & Standalone LLM-as-a-Judge Blindness

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| Model Identifier | Sycophantic Shift (b: P->F) | Precautionary Shift (c: F->P) | Exact McNemar p | Crit
| :--- | :---: | :---: | :---: | :---: | :---: |
```

openai/gpt-oss-120b	0	1	1.0000	4/70 (5.7%)	4/4 (100.0% missed)
gemini-3.1-flash-lite	0	2	0.5000	5/70 (7.1%)	4/5 (80.0% missed)
gemini-3.5-flash-lite	0	1	1.0000	6/70 (8.6%)	6/6 (100.0% missed)
gemma-4-26b-a4b-it	2	0	0.5000	6/70 (8.6%)	6/6 (100.0% missed)
qwen/qwen3.8-27b	4	1	0.3750	11/70 (15.7%)	11/11 (100.0% missed)
openai/gpt-oss-20b	3	5	0.7266	21/70 (30.0%)	7/21 (33.3% missed)
Pooled Cohort (420 Evals)	9	10	1.0000	53/420 (12.6%)	38/53 (71.7% [95% CI: 58.4%-82.0%])

3.3 Table 3: Controlled Architectural Ablation & Baseline Comparison (420 Traces: 53 Unsafe, 367 Safe)

| Evaluation Method / Controlled Ablation Tier | Defect Detection Recall (95% Wilson CI) | Specificity on
| :--- | :---: | :---: | :---: | :---: |
| 1. Standalone LLM-as-a-Judge Baseline | 15/53 (28.3% [18.0-41.6]) | 360/367 (98.1% [96.1-99.1]) | 7/367
| 2. Untyped Lexical Keyword Matcher (No Negation) | 46/53 (86.8% [75.2-93.5]) | 190/367 (51.8% [46.7-56.
| 3. Hybrid Ablation: Omission Rules Only (must_include) | 49/53 (92.5% [82.1-97.0]) | 367/367 (100.0% [9
| 4. Hybrid Ablation: Layer 1 State + Assertion Gate | 53/53 (100.0% [93.2-100.0]) | 367/367 (100.0% [99.
| 5. Full Hybrid Auditor (Layers 1+2+3; v0.2-frozen) | 53/53 (100.0% [93.2-100.0]) | 367/367 (100.0% [99.

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4. Reproducibility & Repository Artifact Map

- Frozen Auditor Engine Tag: v0.2-frozen-rules
- Held-Out Test Scenarios (N=35): study_v1/scenarios/test_scenarios_v1.json
- Strict Evaluation Pipeline: study_v1/run_live_test_evaluation.py
- Statistical & Ablation Scripts: study_v1/run_statistical_analysis.py, study_v1/run_true_ablation.py
- Complete Raw JSON Audit Logs: study_v1/results/live_eval_*.json
- Statistical Summary Artifacts: study_v1/results/final_statistical_summary_v1.json, study_v1/results/ablation_summary_v1.json